

Simulation-Based Development of an Autonomous Ground Vehicle

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Abstract

Autonomous ground vehicles require extensive testing to evaluate vehicle dynamics, navigation performance, and controller behavior under varying operating conditions. While physical testing remains essential, it can be costly, time-intensive, and difficult to reproduce consistently during early stages of development. This project presents the development of a modular digital twin of MARVIN, an autonomous tracked ground vehicle, using MATLAB Simulink and Simscape Multibody to support simulation-based design, validation, and future autonomous system development. The simulation framework separates the vehicle dynamics plant model from the autonomous navigation controller, allowing each subsystem to be developed and validated independently before closed-loop integration. The mechanical model represents the robot's chassis, drivetrain architecture, and tracked vehicle dynamics to estimate vehicle motion and the forces associated with propulsion and maneuvering. Concurrently, a waypoint-based navigation controller generates differential track commands for autonomous path following and mission execution. The modular architecture enables future expansion to incorporate additional subsystems, including sensor models, perception algorithms, and state estimation using LiDAR, cameras, and inertial navigation systems. Upon completion, the digital twin will provide a platform for evaluating vehicle performance, tuning control algorithms, and comparing simulation results with experimental testing on the physical robot. By reducing reliance on repeated field testing and enabling rapid evaluation of design modifications, this framework establishes a scalable foundation for simulation-driven development of autonomous ground vehicles while providing an educational platform for multidisciplinary engineering collaboration.

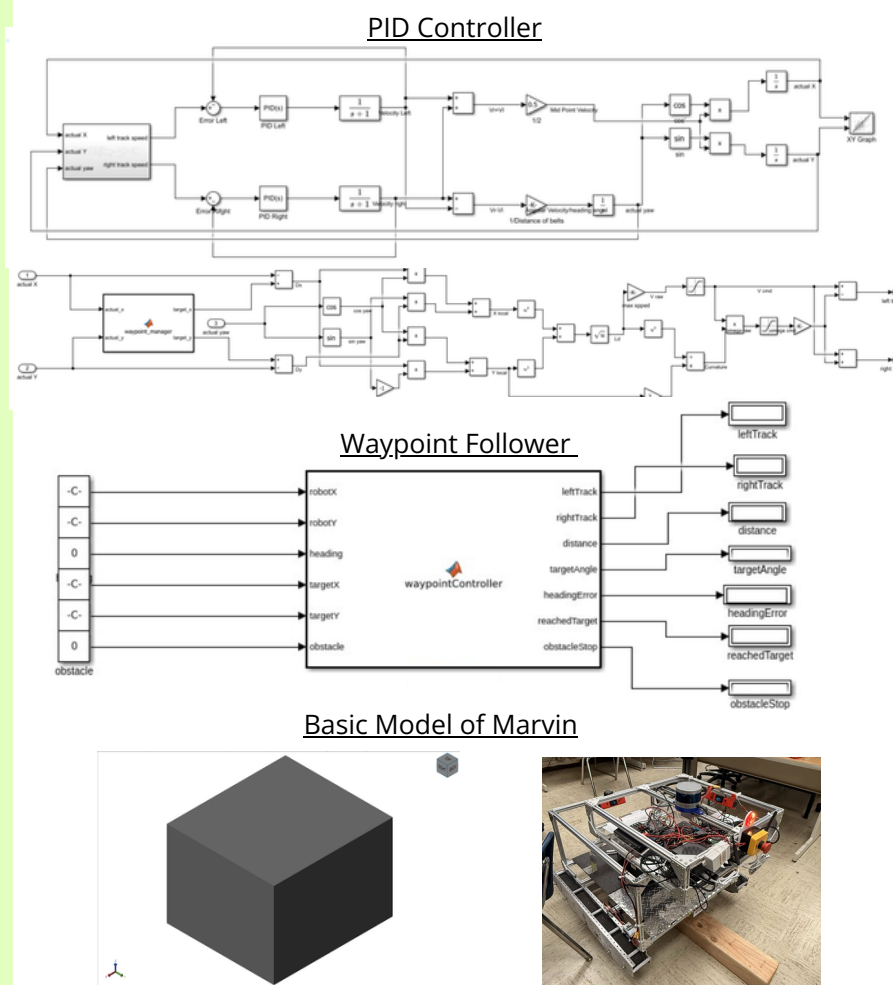
Introduction

Autonomous vehicles require reliable navigation to move safely and accurately through an environment. This project focuses on the simulation-based development of a differential-drive robot capable of navigating to multiple user-defined waypoints. A Pure Pursuit path-tracking algorithm was implemented in MATLAB/Simulink to generate smooth trajectories, while PID controllers regulated the left and right track speeds. The objective was to develop and validate an accurate and robust autonomous navigation system before deployment on a physical platform.

MARVIN is a tracked ground vehicle controlled by independent left and right tracks. A basic three-dimensional vehicle dynamics model is being developed in MATLAB Simulink and Simscape Multibody to simulate forward and backward motion, turning, velocity, position, and yaw rotation. The model uses MARVIN's mass and dimensions to create a simplified representation of its movement. The objective is to establish a foundation that can later be expanded to show how motor torque and track forces affect MARVIN's translational and rotational motion.

Methodology

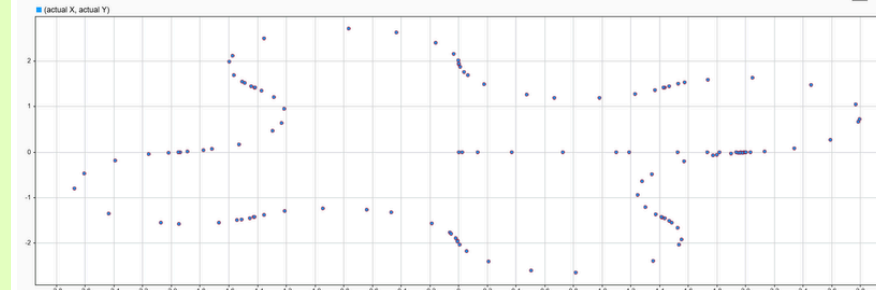
- Learned and applied MATLAB and Simulink to develop and simulate autonomous navigation algorithms.
- Created a PID control system that converts navigation commands into independent left and right track outputs for differential-drive motion.
- Implemented a waypoint navigation subsystem capable of processing multiple waypoints, calculating navigation metrics (distance, target heading, and heading error), and providing real-time outputs for robot guidance.
- Created simplified 3D model of MARVIN using given mass and dimensions.
- Implemented independent left and right track inputs using differential-drive relationships.



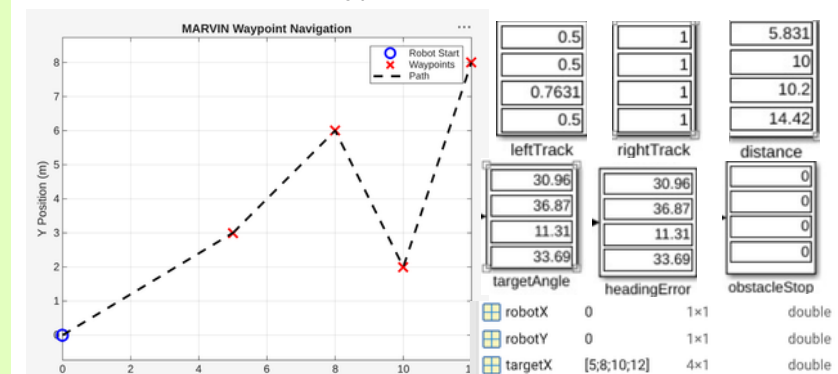
Results

- Successfully validated the PID controller through MATLAB/Simulink simulation
- Successfully validated the waypoint navigation system through graphical simulation and controller outputs.
- Equal track inputs produced straight motion.
- Unequal track inputs produced curved turning motion.
- Opposite track inputs produced pivot rotation.
- Position, velocity, and yaw outputs responded correctly to changes in track commands.
- The model provides a foundation for future torque-based vehicle dynamics analysis.

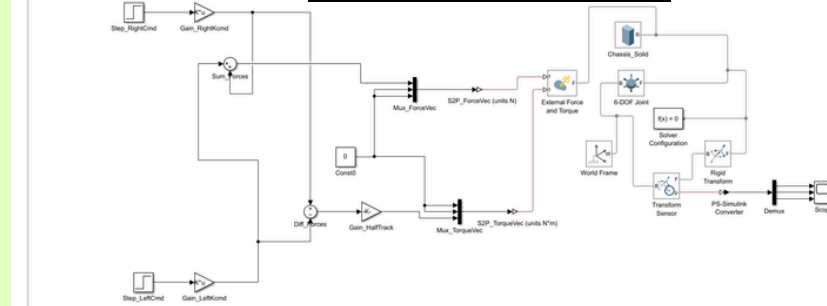
PID Controller Results



Waypoint Follower Results



Marvin Simulation Command



Conclusion

The waypoint navigation and PID control systems for MARVIN were successfully developed and validated in MATLAB/Simulink. A waypoint navigation subsystem was implemented to calculate the distance to each waypoint, determine the target heading and heading error, and generate navigation commands for a user-defined sequence of waypoints. These systems were integrated with a three-dimensional vehicle dynamics model of MARVIN, a tracked ground vehicle controlled by independent left and right tracks, to simulate forward and backward motion, turning, velocity, position, and yaw rotation. By combining differential-drive kinematics, PID speed control, and a Pure Pursuit path-tracking algorithm, the model provides a realistic simulation of MARVIN's navigation and establishes a strong foundation for future hardware implementation, obstacle avoidance, and more advanced autonomous navigation capabilities.

Acknowledgements

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References

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